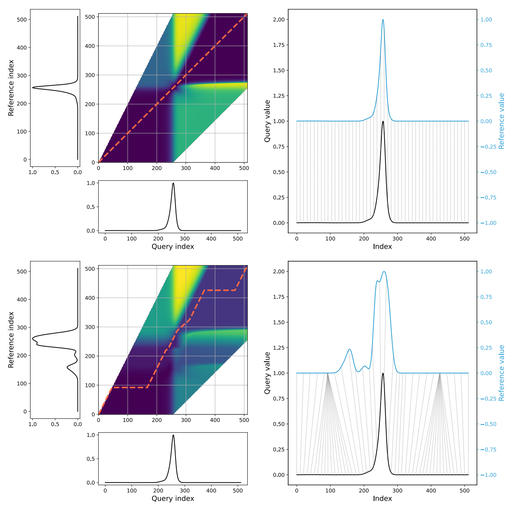
ToPP! Topology of Pulsar Profiles

D. Vohl[1,2,*]

[1]Anton Pannekoek Institute for Astronomy, University of Amsterdam, NL;
[2]ASTRON, Netherlands Institute for Radio Astronomy, NL;
*email: d.vohl@uva.nl

article with J. van Leeuwen & Y. Maan (A&A, 2024)

Comparing profile's Stokes I morphology



$w = 0$ if identical

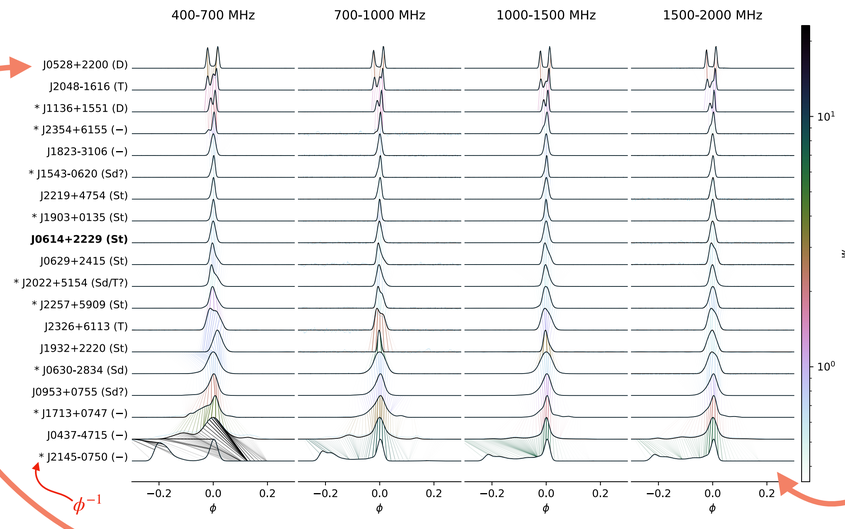
$w > 0$ if different

A **pulsar average profile** (intensity vs rotational phase) is its signature — essential tool for pulsar timing

Average profiles are (mostly) time-stable carrying info about emission regions and geometry

How do 90 pulsars relate?

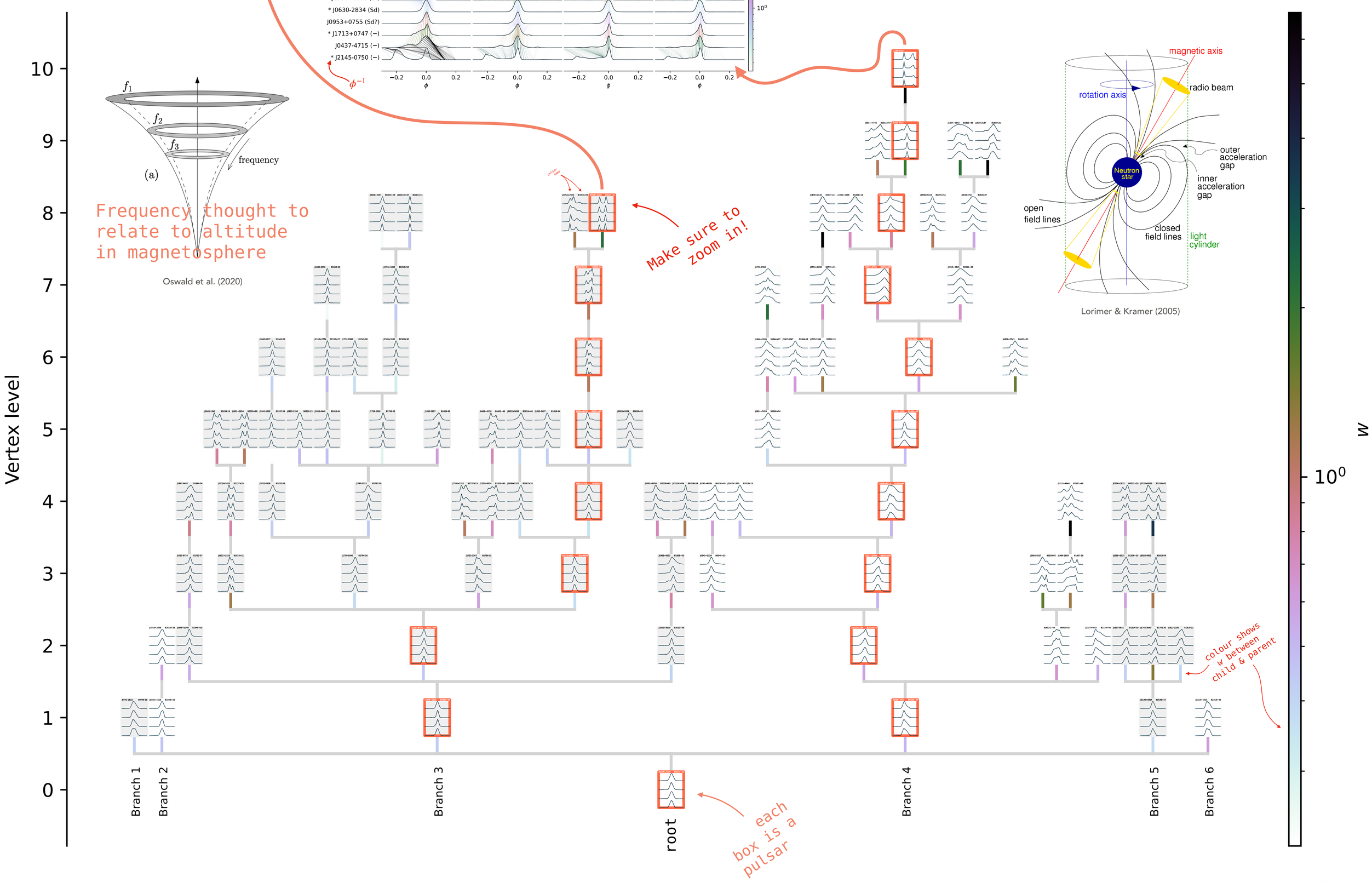
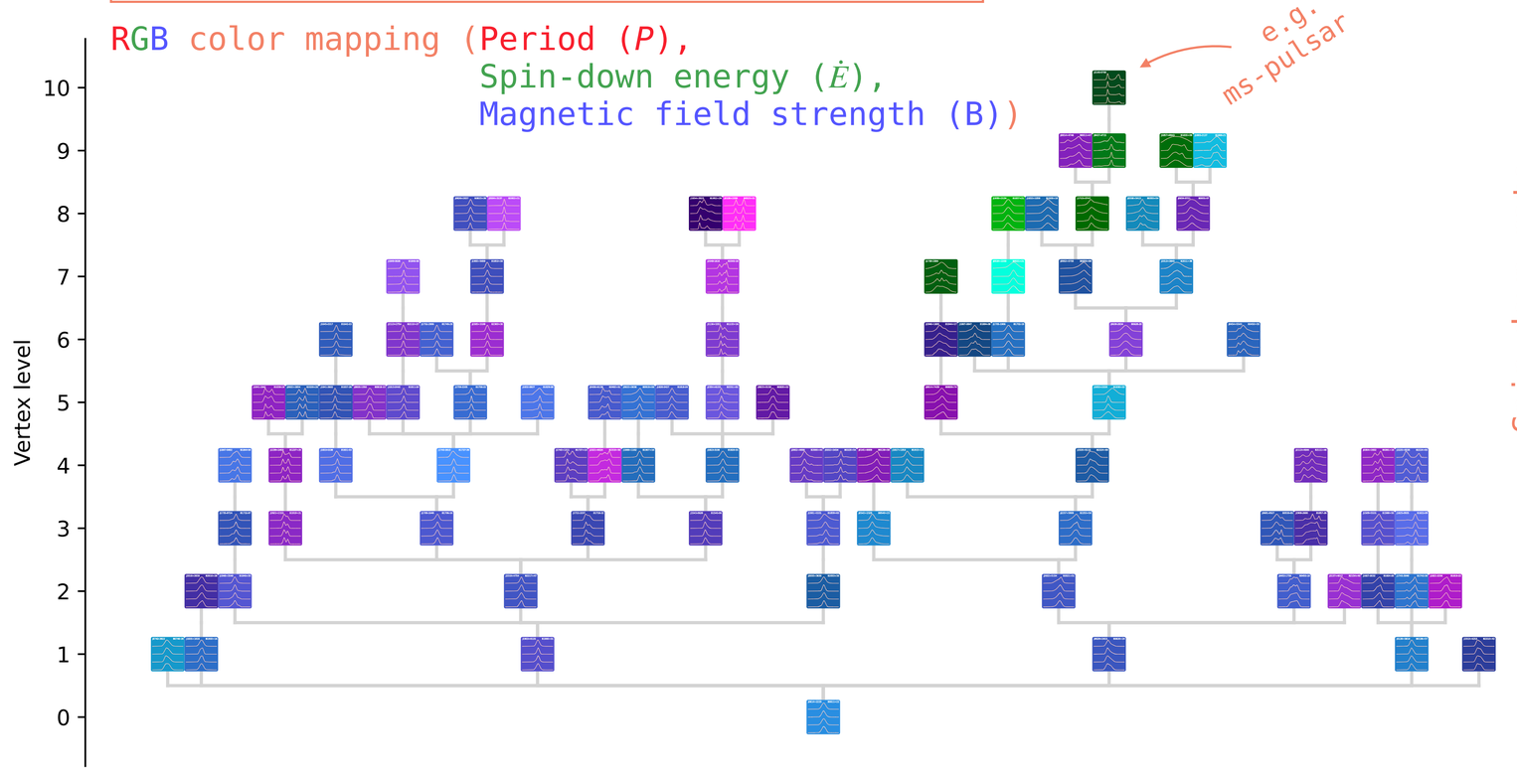
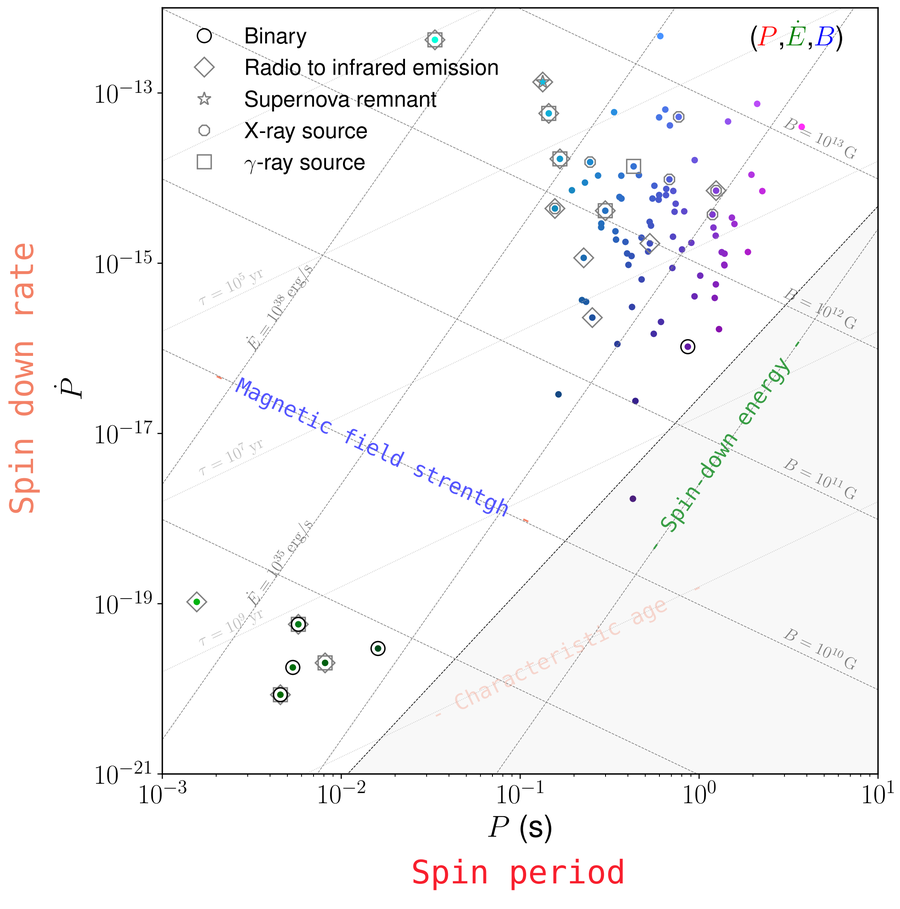
We use graph theory to investigate the European Pulsar Network database.



Sequencing highlights morphology evolution across pulsars

Minimum Spanning Tree on w

(organizes pulsars by spectro-temporal morphology)



We can relate graph regions to physical parameters